

Section 2 Describing Data: Graphical and Numerical

Statistics I · Fall 2026 · Zong-Wei Yeh · Department of Finance, CCU

Overview

Section 1 gave the vocabulary. This section covers the methods of *descriptive* statistics. It shows how to take a set of numbers and say what that set looks like.

There are two ways to do this. The first way is a picture. A picture shows the whole distribution at once: its center, its spread, its shape, its odd values. The second way is a number. A number measures one feature exactly, so that we can compare and compute. Neither way replaces the other. One statistic can hide the shape of a distribution. A picture cannot be put into a formula. So a good description uses both.

This section has four parts.

- §2.1 draws the data. The display follows the type and the structure of the data.
- §2.2 asks where the center of the data is.
- §2.3 asks how spread out the data are.
- §2.4 asks about the shape, and about how two variables move together.

Every measure here is a statistic in the sense of Section 1. It is computed from a sample. Section 7 will treat it as a random variable. Here we only compute it and read it.

2.1 Graphical Methods

Every display begins by counting. We sort the possible values into *classes*. Then we count how many observations fall in each class.

Definition 2.1 (Frequency and relative frequency distribution).

- A frequency distribution (次數分配) is a table of classes with the number of observations (the frequency) in each.
- The relative frequency (相對次數) of a class is its frequency divided by the total number of observations n . Relative frequencies sum to 1.

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Example 2.2. Twenty clients placed the following numbers of trades last month:

7, 12, 2, 5, 16, 9, 10, 6, 14, 4, 8, 11, 15, 5, 13, 3, 7, 18, 10, 9.

Group them into the four classes 0–4, 5–9, 10–14 and 15–19, and build the frequency and relative-frequency distribution. Check the two totals.

A _____ is where a bar chart, a pie chart or a histogram begins. Other displays keep more than counts. A time-series plot keeps the time order, and a scatter plot keeps each pair (x_i, y_i) intact. So the graph to use depends on the type of the data, on its structure, and on what we want to learn from it (Section 1). A categorical variable calls for a bar chart or a pie chart. One numerical variable calls for a histogram. A variable observed over time calls for a line chart. Two variables measured together call for a scatter plot.

Categorical data: bar and pie charts

For a categorical variable the classes are the categories themselves. A *bar chart* (長條圖) draws one bar per category. The height of a bar is the frequency of that category. The bars are _____, because the categories are distinct groups, not adjoining numerical intervals. A *pie chart* (圓餅圖) splits a circle instead. Each slice is proportional to the category's *relative* frequency: a category with relative frequency p gets a slice of _____ degrees. Use a pie chart when the point is each category's share of the whole.

Example 2.3. A survey of 45 investors records the asset class each holds most: Stocks 18, Bonds 12, Funds 9, Cash 6. Display the counts.

Example 2.4. Four brokers split a market with shares 40%, 25%, 20%, 15%. Display the shares.

Numerical data: the histogram

Definition 2.5 (Histogram).

A histogram (直方圖) is the bar graph of the frequency distribution of a numerical variable. The horizontal axis is divided into adjoining class intervals. In this course we always use intervals of equal width. Over each interval we draw a bar whose height is the frequency of that interval. Because the intervals adjoin, the bars _____. In a bar chart they do not.

Why does that matter? The class intervals are adjacent on the numerical scale, so the picture should not show a gap where the scale has none. There is nothing between “just below 0” and

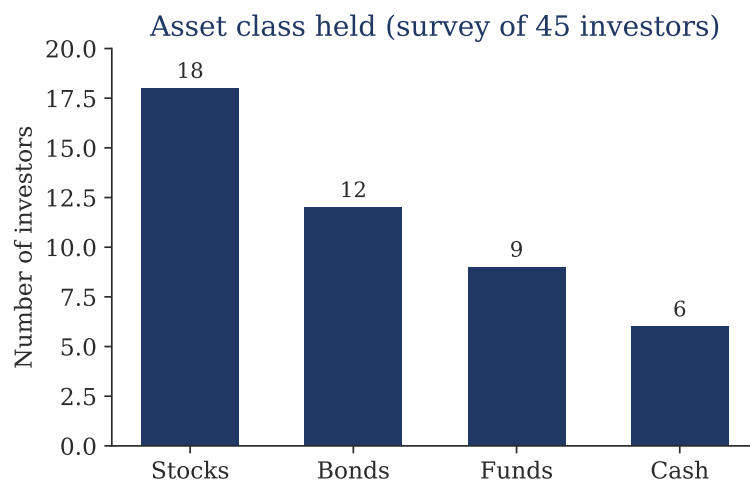


Figure 1: Bar chart of asset class held. The height of each bar is the category's frequency. The categories have no natural order, so we could rearrange them without losing anything. Stocks form the largest category.

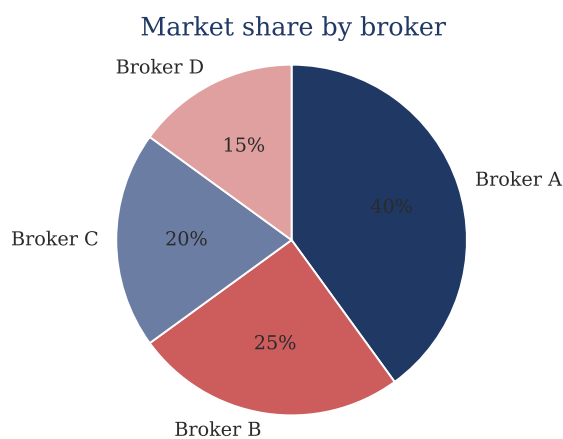


Figure 2: Pie chart of market share. The angle of each slice is proportional to the category's relative frequency, so we read shares of the whole directly. A pie chart works here only because the four shares are parts of one total. It would be meaningless for counts that do not add up to a meaningful whole.

“just above 0”. A histogram also shows center, spread and shape at the same time. That is why we use it more than any other display in this course.

Example 2.6. The daily return of a stock is recorded on 50 trading days and grouped into six classes of width 2%:

class (%)	$[-6, -4)$	$[-4, -2)$	$[-2, 0)$	$[0, 2)$	$[2, 4)$	$[4, 6)$
frequency	3	7	12	14	9	5

Construct the histogram and describe the distribution.

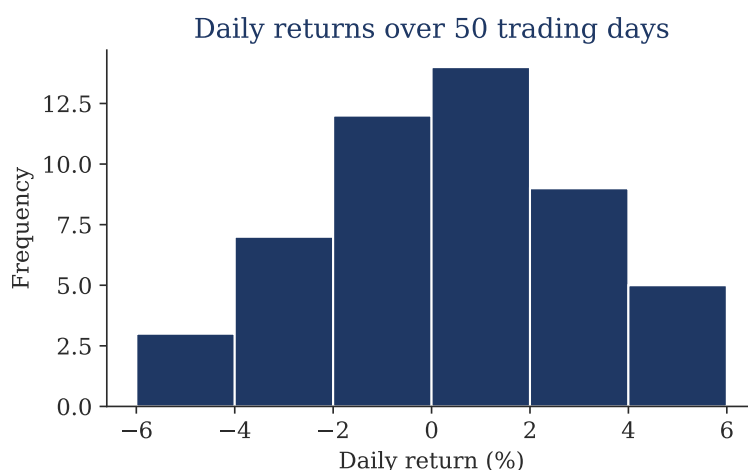


Figure 3: Histogram of 50 daily returns. The relative frequencies are 0.06, 0.14, 0.24, 0.28, 0.18, 0.10, and they sum to 1. The distribution runs from -6% to 6% and is roughly symmetric, centered a little above 0. No single number carries all of that.

Data over time: the line chart

Sometimes we observe the same quantity at successive times. Then we plot the value against time and join the points. This is a *time-series plot* (時間序列圖). Joining the points shows the movement over time. We can see a trend, a seasonal pattern, or a sudden break.

Example 2.7. Monthly sales (in \$000s) of a store for one year (January to December) are

20, 22, 19, 24, 27, 25, 30, 28, 26, 31, 29, 34.

Display the series.

Two variables together: the scatter plot

We often want to know whether two numerical variables move together. To see this, plot each observation as a point (x_i, y_i) . The result is a *scatter plot* (散布圖). It is the picture behind the correlation coefficient of §2.4.

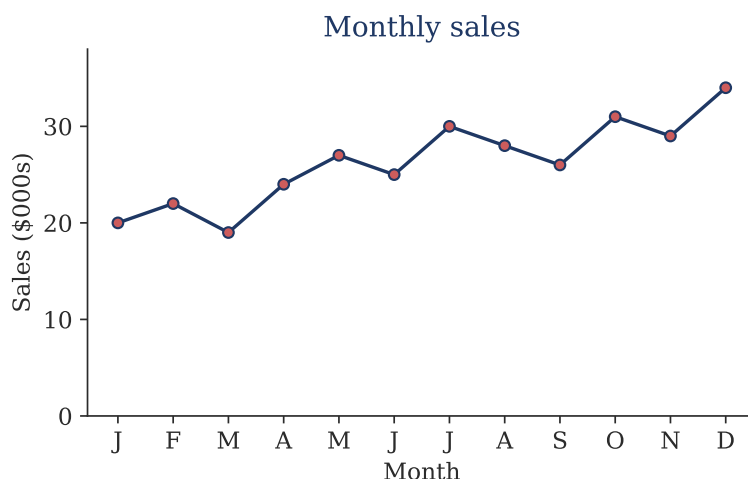


Figure 4: Monthly sales. Joining the points shows what a bar chart of the same numbers would hide. There is a clear upward trend across the year, with small ups and downs around it. Order matters here, so the horizontal axis must stay in time order.

Example 2.8. Five stores report advertising spend x and sales y (both in \$000s):

$$(1, 1), (2, 3), (3, 2), (4, 4), (5, 5).$$

Display the pairs.

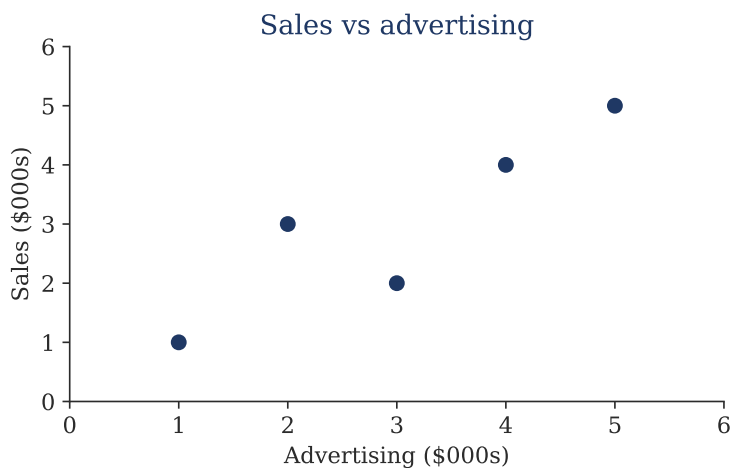


Figure 5: Scatter plot of sales against advertising. The points rise together from lower left to upper right. Higher advertising goes with higher sales. §2.4 turns this picture into a single number, the correlation $r = 0.90$.

The five-number summary and the box plot

A compact numerical picture of a distribution is its *five-number summary* (五數摘要). It lists the minimum, the first quartile Q_1 , the median, the third quartile Q_3 , and the maximum. §2.2 defines the median and the quartiles precisely. Here we only read them off. The *box-and-whisker plot* (盒鬚圖) draws that summary. The box runs from Q_1 to Q_3 , with the median marked inside it.

The whiskers reach the most extreme ordinary values. An extreme value that lies too far out is drawn as a separate point instead. §2.3 gives the rule that decides which values those are.

Example 2.9. The following 11 values (sorted) have five-number summary 12, 15, 19, 22, 40: 12, 14, 15, 16, 18, 19, 20, 21, 22, 24, 40. Display the box plot. (The quartiles and the rule for extreme values are computed in Example 2.26.)

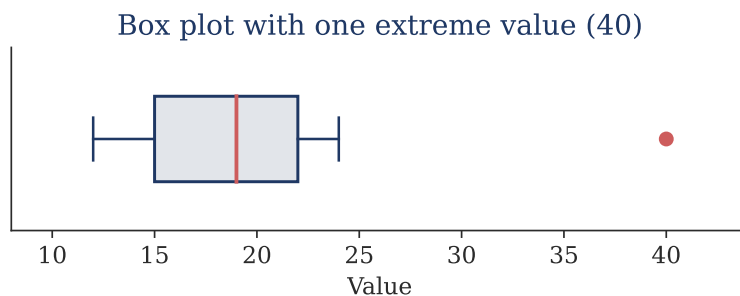


Figure 6: Box plot of the eleven values. The box spans the middle half of the data, from $Q_1 = 15$ to $Q_3 = 22$. The red line is the median, 19. The right whisker stops at the largest ordinary value, 24. The value 40 sits far past the whisker, and it is drawn as a point. Example 2.26 confirms that in §2.3.

2.2 Measures of Location

A measure of location answers one question with one number: where is the distribution? There are several such measures, because “where” has more than one sensible meaning. The measures below need not agree with one another.

Definition 2.10 (Mean, median, mode).

For observations $x_1, \dots, x_n$:

- (i) the mean (平均數) is $\bar{x} =$ _____, the balance point of the data;
- (ii) the median (中位數) is the value in the _____ once the data are ordered; when n is even, average the two central values;
- (iii) the mode (眾數) is the _____ value, and there may be none or several.

The three measures differ in how an extreme value moves them. The mean uses every observation, so one very large value pulls it up. The median depends only on the central values of the ordered data, so it hardly moves. We say that the median is _____. The mode ignores size altogether. When the three measures come apart, the gap between the mean and the median is a useful clue about the shape (§2.4). It is a rule of thumb, not a rule.

Example 2.11. Ten employees earn (in \$000s): 3, 3, 4, 5, 6, 7, 8, 9, 10, 45. Find the mean, median, and mode, and say what they tell you about the shape.

Two special means

Definition 2.12 (Geometric mean).

The geometric mean (幾何平均數) of n positive numbers is

$$\bar{x}_g = \sqrt[n]{x_1 x_2 \cdots x_n}.$$

For a quantity that grows, apply it to the growth factors $(1 + r_i)$. The geometric mean factor is then the constant factor that reproduces the same overall growth.

Returns and growth rates compound, and for them the two means answer different questions. The geometric mean gives the constant per-period growth rate that reproduces the same ending value. The arithmetic mean gives the average one-period return, which is a different quantity. For a multi-period record it is the geometric mean that matches what actually happened to the money.

Example 2.13. \$100 is invested. In year 1 it gains 100% (to \$200); in year 2 it loses 50% (back to \$100). What was the average annual return?

Definition 2.14 (Weighted mean).

When observation x_i carries weight w_i , the weighted mean (加權平均數) is

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}.$$

The ordinary mean is the special case of equal weights. Use the weighted mean whenever the observations stand for pieces of different size: portfolio holdings, course credits, group sizes.

Example 2.15. A portfolio holds 50% of its value in asset A (which returned 8%), 30% in B (12%), and 20% in C (−5%). Find the portfolio return.

Percentiles and quartiles

Definition 2.16 (Percentiles, quartiles, five-number summary).

Order the data from smallest to largest.

- (i) The P th percentile (百分位數) is a value such that about $P\%$ of the observations are _____ it. By the position method it sits in ordered position _____. Interpolate between neighbors when that position is not a whole number.
- (ii) The quartiles (四分位數) are $Q_1 = P_{25}$, $Q_2 = P_{50}$ (the median), and $Q_3 = P_{75}$, at positions _____.
- (iii) The five-number summary is min, Q_1 , median, Q_3 , max.

A percentile reports where a value stands in the data. Only the top tenth of the data exceed the 90th percentile. The three quartiles cut the data into quarters. With the minimum and the maximum they give the five-number summary that the box plot draws.

Example 2.17. For the 11 sorted values 4, 6, 7, 8, 10, 10, 11, 12, 13, 14, 15, find the quartiles and the five-number summary, then the 60th percentile.

Remark (a second convention). Software rarely uses the position method. Excel's PERCENTILE.INC and numpy's default place the P th percentile at rank $1 + \frac{P}{100}(n - 1)$, and interpolate there. For the data above that puts the 60th percentile at rank $1 + 0.60(10) = 7$, so at the 7th value = 11, not 11.2. Both conventions are in everyday use. So always say which one you mean. For small hand computations we use the position method throughout this course.

2.3 Measures of Variability

Two data sets can share a mean and look nothing alike. One can be tightly packed and the other widely spread. A measure of variability puts a number on that spread.

Definition 2.18 (Range and interquartile range).

- (i) The range (全距) is _____.

(ii) The interquartile range (四分位距) is $IQR = \underline{\hspace{2cm}}$, the width of the middle half of the data.

The range uses only the two extremes, so a single extreme value controls it. The interquartile range ignores the outer quarters, so it is resistant. That is why the box plot is built from it.

Definition 2.19 (Variance and standard deviation).

For a sample $x_1, \dots, x_n$ with mean $\bar{x}$, the variance (變異數) and the standard deviation (標準差) are

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

For a whole population of size N with mean μ , divide by N instead: $\sigma^2 = \frac{1}{N} \sum (x_i - \mu)^2$.

The variance averages the squared distances from the mean. We square them so that positive and negative deviations do not cancel. Its square root returns to the original units. That root is the standard measure of spread. Why $n-1$ and not n for a sample? Dividing by n gives the population variance on average. Dividing by $n-1$ corrects that. The full reason waits for Section 7. That section treats a sample statistic as a random variable. The claim can then be proved. Here we simply use $n-1$ whenever we describe a *sample*.

Example 2.20. A fund's monthly returns (%) over five months are 2, 4, 6, 8, 10. Find the range, the sample variance, and the sample standard deviation.

Definition 2.21 (Coefficient of variation).

For data with positive mean, the coefficient of variation (變異係數) is

$$CV = \frac{s}{\bar{x}},$$

usually reported as a percentage. It measures spread relative to the mean, and it is unitless.

The coefficient of variation carries no units. So it compares the variability of series measured in different units or on different scales. The standard deviation on its own cannot do that.

The coefficient of variation is most useful when the mean is positive and well away from zero. A mean near zero makes the ratio explode, and a negative mean makes it hard to read at all.

Example 2.22. Asset X has mean return 8% and standard deviation 12%; asset Y has mean 15% and standard deviation 18%. Which is more variable *relative* to its return?

How much data lies near the mean?**Proposition 2.23** (Chebyshev's theorem).

For any data set and any $k > 1$, at least _____ of the observations lie within k standard deviations of the mean, that is, in the interval $[\bar{x} - ks, \bar{x} + ks]$.

Chebyshev's theorem (柴比雪夫定理) is stated here and proved in Section 4. The proof needs a probability inequality that we meet there. We have no probability yet. The theorem needs *no* assumption about the shape of the data. In return its bound is weak. When the data are approximately normal, meaning bell-shaped and symmetric, the *Empirical Rule* (經驗法則) does much better. It says that about _____, _____ and _____ of the data lie within 1, 2 and 3 standard deviations of the mean.

Definition 2.24 (z-score).

The z-score (標準分數) of an observation x is

$$z = \frac{x - \bar{x}}{s}.$$

It counts how many standard deviations x lies above the mean (if positive) or below it (if negative).

The z-score strips away units and scale. Values from different distributions then become comparable. A z of 2 means “two standard deviations above average”, whether the data are exam scores or returns.

Example 2.25. Exam scores have mean 72 and standard deviation 4. (a) Give intervals guaranteed by Chebyshev to hold at least 75% and at least 88.9% of the scores. (b) If instead the scores are approximately normal, what does the Empirical Rule say for two standard deviations? (c) Find the z-score of a score of 80.

Extreme values and the box plot, revisited

An *outlier* (離群値) is an observation far from the bulk of the data. The standard rule marks off two bounds, called the *fences*:

$$\text{lower fence} = \bar{x} - 1.5s, \quad \text{upper fence} = \bar{x} + 1.5s.$$

An observation outside the fences is flagged as a *potential* outlier and deserves a closer look. It is not automatically an error, and it is not automatically to be dropped. The whiskers of the box plot stop at the most extreme values inside the fences. Anything past them is drawn as a point.

Example 2.26. For the eleven values 12, 14, 15, 16, 18, 19, 20, 21, 22, 24, 40 of Example 2.9, find the quartiles and use the fence rule to identify any outliers.

2.4 Shape and Association

Shape

A distribution has a shape as well as a center and a spread. The most important feature of that shape is *skewness* (偏態), the _____ of the two tails.

A distribution is skewed to the right when it has a long right tail, made by a few large values. It is skewed to the left when it has a long left tail. The gap between mean and median in Example 2.11 was the numerical form of this picture. A mean above the median came from a right tail. A second feature of shape is *kurtosis* (峰態). It describes how often a distribution produces extreme observations, through its tails. We describe kurtosis in words here. Its formula, and a formal skewness coefficient, are deferred to Section 4.3, because both are built from moments that we have not defined yet.

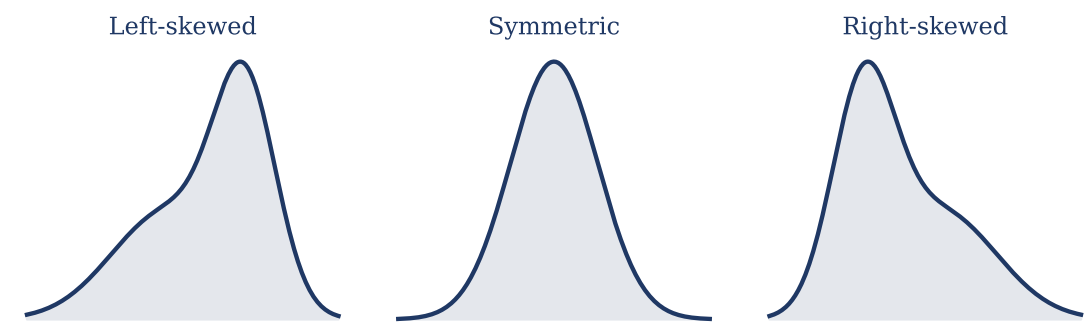


Figure 7: Three shapes. The middle one is symmetric, so its mean and median are about equal. The one on the right has a long right tail, which pulls the mean above the median. The one on the left is its mirror image, with the mean below the median. This is why the mean–median comparison of §2.2 is a useful clue to the three.

Association between two variables

For two numerical variables measured on the same observational units, we ask whether they move together. The scatter plot of Example 2.8 shows the pattern. Two statistics summarize it.

Definition 2.27 (Sample covariance and correlation).

For pairs $(x_1, y_1), \dots, (x_n, y_n)$ with means $\bar{x}, \bar{y}$:

(i) the sample covariance (共變異數) is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) ;$$

(ii) the sample correlation (相關係數) is

$$r = \frac{s_{xy}}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} ,$$

which always satisfies $-1 \leq r \leq 1$.

The covariance gives the _____ of the linear relationship. It is positive when the two variables tend to rise together, and negative when one rises as the other falls. But its size depends on the units. Rescaling dollars to cents multiplies it by 100. So its size means little on its own. Dividing by the two standard deviations removes the units and caps the result in $[-1, 1]$. The correlation r therefore measures the _____ of the linear relationship on a fixed scale. Here ± 1 is a perfect straight line and 0 is no linear association. Two warnings follow. First, r measures *linear* association only. A perfectly deterministic nonlinear relationship can still give $r = 0$: for $y = x^2$ with the x -values symmetric about zero, r is exactly 0. Second, correlation is not causation.

Example 2.28. For the five stores of Example 2.8, with advertising x and sales y equal to $(1, 1), (2, 3), (3, 2), (4, 4), (5, 5)$, find the sample covariance and correlation.

Exercises

Exercise 2.1. A sample of 40 transactions is grouped by size into five classes with frequencies 4, 10, 14, 8, 4. Complete the relative-frequency column and state which class is modal.

Exercise 2.2. For the data 2, 7, 8, 8, 9, 9, 9, 10 find the mean, median, and mode, and state the direction of skew.

Exercise 2.3. A student takes three courses worth 3, 4, and 2 credits and earns grade points 4.0, 3.0, and 2.0. Find the credit-weighted grade-point average.

Exercise 2.4. \$1000 is invested in a fund that returns +30% in year 1 and −30% in year 2. Find the ending value, the arithmetic mean return, and the geometric mean return, and explain the gap.

Exercise 2.5. For the eleven sorted values 4, 6, 7, 8, 10, 10, 11, 12, 13, 14, 15 find the 25th and 90th percentiles by the position method, then by the Excel interpolation method, and comment.

Exercise 2.6. A sample of five returns (%) is 4, 7, 10, 10, 14. Find the mean, the sample variance and standard deviation, and the coefficient of variation.

Exercise 2.7. A portfolio's monthly returns have mean 1.0% and standard deviation 0.5%. Using Chebyshev's theorem, give an interval guaranteed to contain at least 75% of the months, and one for at least 88.9%.

Exercise 2.8. Four observations of two variables are (2, 6), (4, 5), (6, 8), (8, 9). Find the sample covariance and correlation. Explain why the correlation, but not the covariance, can be compared with an association measured in different units.

Exercise 2.9. Two samples of seven observations each:

A : 1, 1, 1, 5, 9, 9, 9

B : 3, 4, 5, 5, 5, 6, 7.

(a) Find the mean and the median of each. (b) Find the mode of each, and say what goes wrong with the mode for A. (c) Find the range of each. (d) Both samples have the same mean and the same median. Describe, in words, how their shapes differ, and say which display would show the difference at once.

Exercise 2.10. (a) Fund P has mean monthly return 0.4% and standard deviation 6%. Fund Q has mean 8% and standard deviation 12%. Compute the coefficient of variation for each, then say whether “P carries ten times the variability of Q per unit of return” is a sound reading. (b) Five observations of two variables are (−2, 4), (−1, 1), (0, 0), (1, 1), (2, 4), so that $y = x^2$ exactly. Compute the sample covariance. What does it say about the relationship, and what does it miss? (c) Now keep only (0, 0), (1, 1), (2, 4). Compute the sample covariance again and explain why the answer changed.

Exercise 2.11. A shop records its daily sales, in thousands of dollars, on ten consecutive trading days, listed in time order:

10, 12, 14, 16, 18, 20, 22, 24, 26, 48.

(a) Find the mean, the median, and Q_1 , Q_3 and the IQR by the position method. (b) Find the two fences of the 1.5 IQR rule and say which observations they flag. (c) The last day was a public holiday with a one-off bulk order, and the figure is correct. Should it be deleted before reporting the average daily sales? Say what you would report instead. (d) An analyst groups the ten numbers into a frequency distribution and draws a histogram, then reports that sales are “roughly symmetric apart from one large value”. What has the histogram thrown away, and which display would have kept it?

Exercise 2.12. *Two funds report monthly returns, in percent, over the same six months:*

Fund A: -1 1 1 3 4 4

Fund B: -2 2 6 2 8 8

- (a) Find the mean and the median of each fund. (b) Find the sample variance and the sample standard deviation of each. (c) Find the coefficient of variation of each, and say which fund is more variable relative to its own average return. (d) Find the sample covariance and the sample correlation between the two funds. (e) A client reads your table and concludes: “B is the better fund, and holding both together removes most of the risk.” Address both halves, and state one thing these six numbers cannot tell you.